

Spectral differentiation: Let $u(x)$ be a smooth periodic function defined on an interval $[0, L]$ for some $L > 0$, where $u(L) = u(0)$. The derivative of $u(x)$ is given by differentiating the Fourier transform of $u(x)$.

$$\frac{\partial u}{\partial t} = k \frac{\partial u}{\partial x} \equiv F(u) \quad \left| \begin{array}{l} \text{Fourier method} \\ \text{tells about the} \\ \text{kernel.} \end{array} \right.$$

$$u(x, t) = \int_{-\infty}^{\infty} \hat{u}(y) \frac{1}{\sqrt{4\pi kt}} e^{-\frac{(x-y)^2}{4kt}} dy = \int_{-\infty}^{\infty} \hat{u}(y) K(x-y) dy$$

To apply discrete Fourier transform on a grid $\{x_j\}$, consider the grid points as $x_0 = -L/2, x_1 = -L/2 + dx, \dots, x_N = L/2 - \Delta x$, where $\Delta x = L/N$. Notice that $x_{N+1} = L/2$ has not been considered. It is recommended that N be some power of 2, although N can be any integer. The wave numbers are defined as

$$k = [0, 1, \dots, N/2 - 1, -N/2, \dots, -1](2\pi/L)$$

$2\pi i k x / L$
e

Let us demonstrate the Fourier spectral differentiation by considering the derivative of

$$u(x) = \cos(x)e^{-x^2/25}$$

The following python code computes the derivative of $u(x)$.

```
n = 128
L = 30
dx = L/n
x = np.arange(-L/2, L/2, dx)
u = np.cos(x) * np.exp(-np.power(x, 2)/25) # Function
du_exact = -(np.sin(x) * np.exp(-np.power(x, 2)/25) + (2/25)*x*f) # Derivative

## Derivative using FFT (spectral derivative)
uHat = np.fft.fft(u)
```

$k = 2\pi \cdot \text{np.pi} \cdot \text{np.fft.fftfreq}(n, d=dx)$

 $\xrightarrow{i k \hat{u}}$

```
duHat = k * uHat * (1j)
du = np.real(np.fft.ifft(duHat))
plt.plot(x, du)
plt.plot(x, du_exact)
```

$\frac{\partial u}{\partial x}$

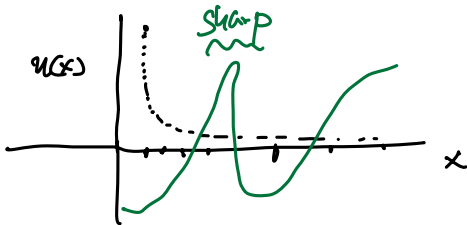
Number of Fourier modes detects the dimension of the system.

Remark: If $u(x)$ has sharp change near $x = \bar{x}$

then Fourier method may show wiggles.

Runge phenomena or Gibbs oscillation.

Spectral differentiation on non-uniform grids: In boundary layer problems, steep gradients develop near solid surfaces. Accurate numerical resolution of these features requires grid clustering near the boundaries. This type of grid clustering is called adaptive grids.



In FD: $\frac{\partial u}{\partial x} = \frac{u_{j+1} - u_j}{x_{j+1} - x_j}$

if $\Delta x_j = x_{j+1} - x_j$ is not constant.

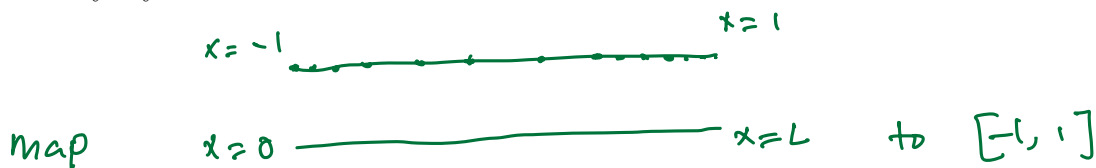
One example of an adaptive grid is in which the grid spacing $\Delta x_j = x_{j+1} - x_j$ is nonuniform. Unfortunately, standard finite difference methods lose accuracy on such grids due to truncation errors. In contrast, accurate spectral differentiation on a non-uniform grid can be achieved by using interpolation methods to approximate differential operators.

Goal: find a polynomial $p(x)$ that fits u_j on a non uniform grid. such that $p(x_j) = u(x_j) = u_j$.
differentiate $p(x)$ to find $\frac{\partial u}{\partial x} \Rightarrow \underline{p'(x_j) = u'(x_j)}$

Spectral methods can be relatively very accurate on nonuniform grids if we use a suitable interpolator leading to a differentiation matrix. To understand this method, consider a one-dimensional adaptive grid defined by

$$x_j = \cos(j\pi/N), \quad j = 0, 1, \dots, N, \quad x \in [-1, 1]$$

which clusters the grid as needed near the boundaries $x = \pm 1$, making it ideal for resolving boundary layers.



For a function $u(x)$ evaluated at grid points $\{x_j\}$ of the interval $[-1, 1]$, let $\mathbf{u} = [u_0, u_1, \dots, u_N]^T$ be the vector containing function values at the grid points such that $u_j = u(x_j)$.

Algorithm for spectral differentiation: Consider a unique polynomial $p(x)$ of degree at most N such that $p(x_j) = u_j$ for $j = 0, 1, \dots, N$, which interpolates $u(x)$ for all $x \in [-1, 1]$. Differentiating this polynomial and evaluating at grid points, we get the vector $\mathbf{p}' = [p'(x_0), p'(x_1), \dots, p'(x_N)]^T$. Since $p(x)$ interpolates $u(x)$ at grid points, $\mathbf{p}'$ is a vector that also approximates the derivative $u'(x)$ on each grid points $\{x_j\}$.

Want to find a transformation between $\mathbf{p}'$ and $\mathbf{u}'$

In finite difference method, numerical approximation of derivatives are associated with a finite difference matrix. Similarly in spectral method, there exists an $(N+1) \times (N+1)$ matrix $\mathcal{D}$ such that the vector $\mathcal{D}\mathbf{u} \approx [p'(x_0), p'(x_1), \dots, p'(x_N)]^T$, i.e., the product $\mathcal{D}\mathbf{u}$ provides the spectral approximation of the derivative of $u(x)$ on the adaptive grid $\{x_j\}$.

$\mathcal{D}$ is called spectral differentiation matrix.

Simple example: Below we consider two examples illustrating the construction and use of spectral differentiation matrices.

1. Consider a grid $\{-1, 1\}$ with only two points. Let $u_0 \equiv u(1)$ and $u_1 \equiv u(-1)$ be the values of $u(x)$ on this grid. Construct an interpolant:

$$p(x) = \frac{1}{2}(1+x)u_0 + \frac{1}{2}(1-x)u_1.$$

Observe that

$$u_0 = p(1) \quad \text{and} \quad u_1 = p(-1).$$

Moreover,

$$p(x) = \sum_{j=0}^1 u_j p_j(x),$$

where $p_0(x) = (1/2)(1+x)$ and $p_1(x) = (1/2)(1-x)$.

e.g. $p_0(x) = 1+x+x^2$??

$$\begin{cases} p'(x) = \frac{1}{2}u_0 - \frac{1}{2}u_1 \\ p'(-1) = \frac{1}{2}u_0 - \frac{1}{2}u_1 \\ p'(1) = \frac{1}{2}u_0 - \frac{1}{2}u_1 \end{cases}$$

$$\begin{bmatrix} p'(-1) \\ p'(1) \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{bmatrix} \begin{bmatrix} u_0 \\ u_1 \end{bmatrix}$$

Differentiating $p(x)$ with respect to x , we obtain the differentiation matrix:

$$\mathcal{D} = \begin{pmatrix} \frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{pmatrix} \quad \checkmark$$

We see that $\mathcal{D}\mathbf{u}$ gives $[p'(x_0), p'(x_1)]^T$, where $x_0 = 1$ and $x_1 = -1$.

2. Consider a 3-point grid $\{-1, 0, 1\}$. Given an interpolant

$$p(x) = \frac{1}{2}x(1+x)u_0 + (1+x)(1-x)u_1 + \frac{1}{2}x(x-1)u_2.$$

Write this interpolant as

$$p(x) = \sum_{j=0}^2 u_j p_j(x),$$

and identify the form of $p_j(x)$ for $j = 0, 1, 2$. Also, find the 3×3 differentiation matrix.

Find $\mathcal{D}$ (H.W)

Software: Let $p_j(x)$ be N -degree polynomial interpolant to the delta function supported at x_j grid point, i.e. $p_j(x_i) = \delta_{ij}$, which interpolates a data vector $\{u_j\}$ as

$$u(x) = \sum_{j=0}^N u_j p_j(x).$$

Differentiating the polynomial $p_j(x)$ with respect to x and evaluating at x_j gives the value of $\frac{\partial u}{\partial x}$ at grid point x_j . Let us take $p'_j(x)$ and evaluate it at x_i for $i = 0, \dots, N$ and form a vector $[p'_j(x_i)]^T$. If we collect these vectors for $j = 0, \dots, N$, then we get $N+1 \times N+1$ matrix $\mathcal{D}$, given by

No need to
know how to
derive this

$$\begin{aligned} D_{00} &= \frac{2N^2 + 1}{6} \\ D_{jj} &= \frac{-x_j}{2(1 - x_j^2)}, \quad j = 1, \dots, N-1 \\ D_{NN} &= -D_{00} \\ D_{ij} &= \frac{c_i(-1)^{i+j}}{c_j(x_i - x_j)}, \quad i \neq j, \quad i, j = 0, \dots, N \end{aligned}$$

where $c_i = 2$ for $i = 0$ or N , otherwise $c_i = 1$.

```
def spectral_diff(N):
    """
    Return (D, x) where:
        x: points on [-1,1], size N+1
        D: first-derivative matrix on those points, shape (N+1,N+1)
    """
    if N == 0:
        return np.array([[0.0]]), np.array([1.0])

    x = np.cos(np.pi * np.arange(N+1) / N).T

    c = np.ones(N+1)
    c[0] = 2.0
    c[-1] = 2.0
    c = c * ((-1.0) ** np.arange(N+1))

    X = np.tile(x, (N+1, 1)).T
    dX = X - X.T

    D = (np.outer(c, 1/c)) / (dX + np.eye(N+1))
    D = D - np.diag(np.sum(D, axis=1))
    return D, x
```

Remark:

- 1) use FFT to find $\frac{\partial u}{\partial x}$
- 2) use spectral diff code to find $\frac{\partial u}{\partial x}$

Eigenvalues and spectrum of matrices and operators: The spectral radius of $N \times N$ matrix A is defined by

$$\rho(A) = \max |\lambda_k|$$

where λ_k is a solution of $|A - \lambda I| = 0$.

The condition number of A is $\kappa(A) = \|A\| \|A^{-1}\|$. A matrix A is well condition if the condition number is close to one. Further details are given in Appendix C (Leveque).

Let λ_k be the eigenvalues of a non-defective square matrix A . This matrix has the following decomposition:

$$A = \Phi \Lambda \Phi^{-1},$$

where $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_N)$ and columns of Φ are corresponding eigenvectors.

$$A = \begin{bmatrix} \vdots & \vdots & \dots & \vdots \\ \phi_1 & \phi_2 & \dots & \phi_n \\ \vdots & \vdots & \dots & \vdots \end{bmatrix} \begin{bmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \dots & \\ & & & \lambda_n \end{bmatrix} \begin{bmatrix} \vdots & \vdots & \dots & \vdots \end{bmatrix}$$

We may rewrite this as

$$\Phi^{-1} A = \Lambda \Phi^{-1}$$

which suggests that the rows of Φ^{-1} are left eigenvectors of A .

Let $A \in \mathbb{R}^{m \times n}$ be a real matrix. There exists orthogonal matrices $\Phi \in \mathbb{R}^{m \times m}$, $\Psi \in \mathbb{R}^{n \times n}$, and $\Sigma = \text{diag}(\sigma_1, \dots, \sigma_p) \in \mathbb{R}^{m \times n}$ with $\sigma_1 > \sigma_2 > \dots$ and $p = \min(m, n)$ such that

$$A = \Phi \Sigma \Psi^T.$$

This is called singular value decomposition. (np.linalg.svd(A) => svd)

$$A A^T = \Phi \Sigma \Psi^T (\Phi \Sigma \Psi^T)^T = \Phi \Sigma \Psi^T \Psi \Sigma^T \Phi^T = \Phi \Sigma \Sigma^T \Phi^T, \quad \Psi^T \Psi = I \\ = \Phi \Sigma^V \Phi^T$$

The best rank- r approximation of A is given by

$$\text{Also } A^T A = \Psi \Sigma^T \Psi^T$$

$$A \approx A_r = \sum_{j=1}^r \sigma_j \phi_j \psi_j^T$$

which minimizes $\|A - A_r\|$.

The pseudo inverse of A is given by

$$A^\dagger = \Psi \Sigma^\dagger \Phi^T.$$

$$\text{For } A \in \mathbb{R}^{n \times n}, \quad A A^{-1} = I$$

$$\text{For } A \in \mathbb{R}^{n \times m}, \quad A A^\dagger = I$$

Eigen decomposition example: Find the eigenvalues and eigenvectors of

$$-\frac{\partial^2 u}{\partial x^2} = \lambda u, \quad u(0) = 0 = u(1).$$

Exact solutions are $\phi_n(x) = \sin(n\pi x)$ and $\lambda_n = (n\pi)^2$, $n \geq 1$.

Using second order finite difference method, we can construct A and find the eigen decomposition.

Using the spectral differentiation, we can construct A and compare the eigenvalues by two methods.

$$\text{Convert POE} \rightarrow Au = \lambda u$$

Dynamic modes of PDEs: For linear PDEs

$$\frac{\partial u}{\partial t} = \mathcal{L}u$$

subject to boundary conditions.

e.s. $\mathcal{L}u = -\alpha \frac{\partial u}{\partial x} + \alpha \frac{\partial^2 u}{\partial x^2}$
 $\mathcal{L}u = \kappa \frac{\partial^2 u}{\partial x^2}$

Convert $\mathcal{L} \equiv A$

The dynamic modes of the PDE are eigenfunctions of $\mathcal{L}$:

$$\mathcal{L}\phi_k(x) = \lambda_k \phi_k(x).$$

The dynamic modes span the solution of the PDE as

$$u(x, t) = \sum_k a_k \phi_k(x) e^{\lambda_k t}.$$

$$\boxed{\frac{du}{dt} = Au}$$

$$u = e^{At} u_0$$

$$= \sum a_k \phi_k(x) e^{\lambda_k t}$$

For heat equation with Dirichlet boundary conditions on $[0, 1]$, the dynamic modes are $\phi_n(x) = \sin(n\pi x)$ and eigenvalues are $\lambda_n = -(n\pi)^2$. These are analytically found by method of separation of variables.

In a previous lecture, we have seen that forward time and centered space finite difference approximation of heat equation leads to a linear system

$$U^{n+1} = AU^n$$

← numerical PDE $\Rightarrow \omega$

where U^n denotes the solution vector at grid points at time t_n . This linear system can be generalized for any linear PDE

$$\frac{\partial u}{\partial t} = \mathcal{L}u$$

← Actual PDE $\Rightarrow \lambda$

Q. $e^A = ?$, let $A = \Phi \Lambda \Phi^T \Rightarrow e^A = \Phi e^\Lambda \Phi^T$

First, we are interested in connecting the eigenvalues of A with eigenvalues of $\mathcal{L}$. Using matrix exponential, we see that

$$U^{n+1} = A U^n$$

$$u(x, t + \Delta t) = e^{\Delta t \mathcal{L}} u(x, t)$$

$$\Rightarrow A = e^{\Delta t \mathcal{L}}$$

Therefore, the exact amplification factor for the dynamic mode is

$$\omega = e^{\lambda \Delta t}.$$

In other words,

$$\lambda = \frac{1}{\Delta t} \ln \omega.$$

λ is eigenvalue of PDE
 ω is eigenvalue of numerical PDE

Second, we are interested in constructing dynamics modes of A and use that to form the solution of the PDE. It is apparent from the FD scheme that

$$U^n = A^n U^0.$$

The question of computational stability is related to the properties of the discretization matrix A .

When the eigenvectors of A form a complete linearly independent set, any given initial condition U^0 can be decomposed as

$$U^0 = \sum_k a_k \phi_k,$$

where a_k are coefficients to be determined. We see that

$$U^n = \sum_k a_k \lambda_k^n \phi_k.$$

End of Spectral method.

Lecture #14, Oct 30, 2025

Finite volume method for conservation law. (chap 4)

this lecture is on conservation law.

A PDF copy of the book has been uploaded.

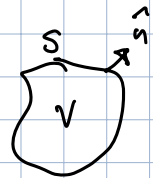
Two goals: basic schemes and limitation
How to improve accuracy.

let $\vec{u}(x,t)$ be a vector and $\vec{F}(\vec{u}) = \phi(\vec{u}) \vec{u}$

① $\frac{\partial \vec{u}}{\partial t} + \nabla \cdot \vec{F} = 0$ is called conservation law.

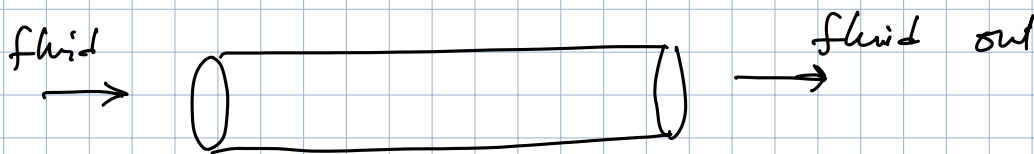
$$\int_V \frac{\partial \vec{u}}{\partial t} dV + \int_V \nabla \cdot \vec{F} dV = 0, \quad V \subseteq \mathbb{R}^3$$

$$\frac{d}{dt} \int_V \vec{u} dV + \iint_S \vec{F} \cdot \hat{n} dS = 0$$



If $\vec{F} \cdot \hat{n} = 0$ then $\frac{d}{dt} \int_V \vec{u} dV = 0$

Thus $\vec{F} \cdot \hat{n}$ is flux normal component of flux vanishes on boundary.



Simple form of ① is

$$\textcircled{2} \quad \frac{\partial u}{\partial t} + \frac{\partial f(u(x,t))}{\partial x} = 0$$

$$\frac{\partial u}{\partial t} + \frac{\partial f}{\partial u} \frac{\partial u}{\partial x} = 0$$

$$\textcircled{3} \quad \frac{\partial u}{\partial t} + a \frac{\partial u}{\partial x} = 0 \quad \text{where } a = \frac{df}{du}$$

$$\text{e.g. } f = u^3, \quad a = \frac{df}{du} = 3u^2$$

(3) has solⁿ $u(x,t) = f(x-at)$, where $u(x,0) = f(x)$

Remark: $\rightarrow$ given flux $f(u)$ What is wave speed?

$\rightarrow$ given a PDE, does it describe conservation law and if so under what condition?

e.g. $\frac{\partial u}{\partial t} = \alpha \frac{\partial^3 u}{\partial x^3}, \quad \alpha > 0$

$$\frac{\partial u}{\partial t} + \frac{d}{dx} \left(-\alpha \frac{\partial^2 u}{\partial x^2} \right) = 0, \quad f(u) = -\alpha \frac{\partial^2 u}{\partial x^2}$$

is flux.

Method of characteristic

$$a \frac{\partial u}{\partial t} + b \frac{\partial u}{\partial x} = c \quad \text{model PDE}$$

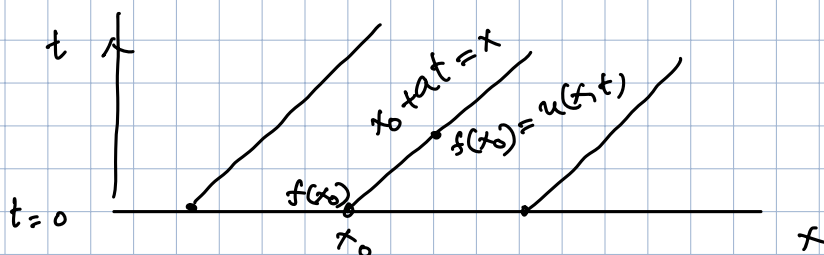
Think of $\frac{dt}{a} = \frac{dx}{b} = \frac{du}{c}$

$$\frac{dx}{dt} = \frac{b}{a}, \quad \frac{du}{dt} = \frac{c}{a}$$

Take $\frac{\partial u}{\partial t} + a \frac{\partial u}{\partial x} = 0$

$$\boxed{\frac{dx}{dt} = a, \quad \frac{du}{dt} = 0}$$

$\Rightarrow u(x,t)$ is constant on $x(t) = at + x_0$
ie. on characteristic line.



$$\text{At } t=0, u(x,0) = f(x) \quad \left\{ \begin{array}{l} u(x,0) = f(x_0) \\ \text{At } t=0, x = a \cdot 0 + x_0 = x_0 \end{array} \right.$$

$$\text{At } t>0, u(x,t) = f(x) \text{ on the line}$$

$$= f(x_0)$$

$$u(x,t) = f(x-at)$$

Exo

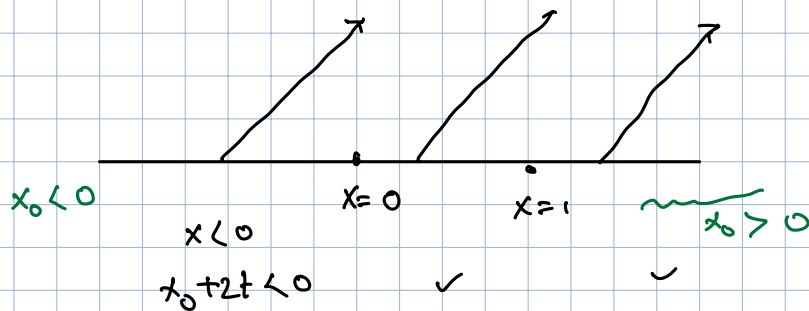
$$\frac{\partial u}{\partial t} + 2 \frac{\partial u}{\partial x} = 0$$

$$u(x,0) = \begin{cases} 0 & x < 0 \\ 4x & 0 < x < 1 \\ 0 & x > 1 \end{cases}$$

find $u(x,t)$

Solⁿ

$$\frac{dx}{dt} = 2, \quad \frac{du}{dt} = 0, \quad x(t) = x_0 + 2t$$



$$4x_0 = 4(x-2t) = 4x - 8t$$

$$u(x,t) = \begin{cases} 0 & x/t > 2, \quad x-2t > 0, \quad x_0 > 0 \\ 4x-8t & 2t < x < 2t+1, \quad 0 < x_0 < 1 \\ 0 & x/t < 2, \quad x-2t < 0, \quad x_0 < 0 \end{cases}$$

Exo:

$$\frac{\partial u}{\partial t} + 3t^v \frac{\partial u}{\partial x} = 2t u$$

$$u(x,0) = f(x)$$

Sol:

$$\frac{dx}{dt} = 3t^v = \text{speed}$$

$$x(t) = t^3 + x_0$$

$$x_0 = x - t^3$$

$$\frac{du}{dt} = 2t u, \quad \frac{du}{u} = 2t dt$$

$$\ln u = t^v + \text{Const}$$

$$u = c e^{t^v}$$

$$u(x,0) = c e^0 = c$$

$$u(x_0,0) = f(x_0) = c$$

$$u(x,t) = f(x_0) e^{t^2} \\ = f(x-t^3) e^{t^2}$$

Ex: $\frac{\partial u}{\partial t} + \frac{\partial}{\partial x} \left(\frac{u^6}{6} \right) = 0$

$$u(x,0) = \begin{cases} 3 & x < 0 \\ 4 & x > 0 \end{cases}$$

Solⁿ

$$\frac{\partial u}{\partial t} + \frac{\partial f}{\partial x} = 0, \quad f = \frac{u^6}{6}$$

$$\frac{\partial u}{\partial t} + \frac{\partial f}{\partial u} \frac{\partial u}{\partial x} = 0$$

$$\frac{\partial u}{\partial t} + u^5 \frac{\partial u}{\partial x} = 0$$

$$\frac{dx}{dt} = u^5, \quad \frac{du}{dt} = 0, \quad u(x,t) \text{ depends on } x \text{ and } t$$

$$\frac{dx}{dt} = (u(x,t))^5$$

$$x(t) = u^5 t + x_0$$

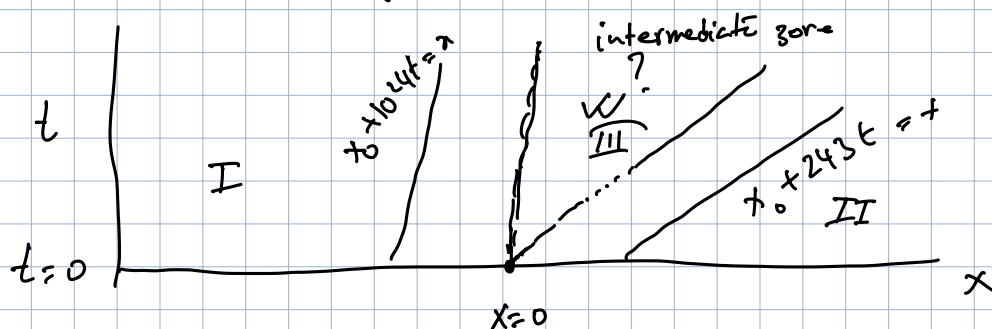
because u is constant on characteristic curve.

For $x < 0$, $u(x,0) = 3$

$$x(t) = 3^5 t + x_0 = x_0 + 243 t$$

For $x > 0$, $u(x,0) = 4$

$$x(t) = 4^5 t + x_0 = x_0 + 1024 t$$

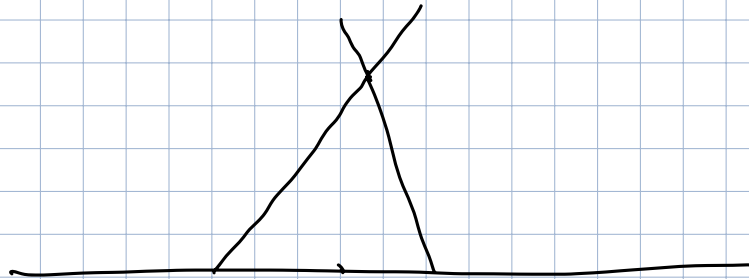


In III, $x = x_0 + u^5 t$ but $x_0 = 0$
 $x = 0 + u^5 t \Rightarrow u = \left(\frac{x}{t} \right)^{1/5}$

$$u(x,t) = \begin{cases} 3 & x/t < 2/3 \\ (x/t)^{1/5} & 2/3 < x/t < 10/24 \\ 4 & x/t > 10/24 \end{cases}$$

Remark: If $u(x,0) = \begin{cases} 4 & x < 0 \\ 3 & x > 0 \end{cases}$

Sketch of curve is :



Characteristic curve intersect

Goal: Find if characteristic curves intersect or not.

System of PDE: (hyperbolic)

$$\frac{\partial u}{\partial t} + A \frac{\partial u}{\partial x} = 0, \quad u(x,t) : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}^n$$

Ex:

$$\begin{cases} \frac{\partial p}{\partial t} + k_0 \frac{\partial u}{\partial x} = 0 \\ \frac{\partial u}{\partial t} + \frac{1}{\rho_0} \frac{\partial p}{\partial x} = 0 \end{cases} \quad \left\{ \begin{array}{l} \frac{d}{dt} \begin{bmatrix} p \\ u \end{bmatrix} + \begin{bmatrix} 0 & k_0 \\ \frac{1}{\rho_0} & 0 \end{bmatrix} \begin{bmatrix} p \\ u \end{bmatrix} \end{array} \right\}_x$$

$\nearrow$
 u

$\nearrow$
 A

$\nearrow$
 $\frac{\partial u}{\partial x}$

what is hyperbolic?

If A has real eigenvalues
and a corresponding set of
linearly independent eigen vectors,
then the system is hyperbolic.

that means

$$A = \Phi \Lambda \Phi^{-1}$$

$$\text{let } q = \Phi^{-1} u$$

$$\text{then } \Phi^T \frac{\partial u}{\partial t} + \Phi^T A \frac{\partial u}{\partial x} = 0$$

$$\frac{\partial q}{\partial t} + \Phi^T A \Phi \Phi^{-1} \frac{\partial u}{\partial x} = 0$$

$$\boxed{\frac{\partial q}{\partial t} + \Lambda \frac{\partial q}{\partial x} = 0}$$

$$\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$$

$$\text{or } \frac{\partial q_k}{\partial t} + \lambda_k \frac{\partial q_k}{\partial x} = 0, \quad q = \begin{bmatrix} q_1 \\ \vdots \\ q_n \end{bmatrix}$$

$$\text{characteristics } \frac{dx}{dt} = \lambda_k$$

$$x(k) = x_0 + \lambda_k t$$

